

Unit 111 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 55.7 ac. 27 plots, BAF 20, variable-radius cruise on a 300 ft grid, 300 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	222 sq ft/ac (1 SE of mean, BA 3.5 %)	Type	Sierran mixed conifer
Trees per acre	110	CWHR size / density	4 / D (cover 86 %)
QMD	19.2 in	SDI	296 (51 % of max 586)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	216 sq ft/ac 20.9 in	BA QMD	7 sq ft/ac 7.9 in
TPA	91	TPA	20
Volume	32.2 MBF/ac (6422 cu ft)	Volume	2.0 green tons/ac (87 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 154 sq ft/ac BA	Vol. sampling error	13.3 % at 95 % (CV 34 %, t = 2.06, n = 27)
Species (BA)	WF 71, PP 63, DF 36, IC 35, SP 18	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 6
		Sale volume share	3.4 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	7.1 % at 95 % (CV 18 %); reference, not the standard
		Measured trees	PP 85, WF 96, DF 48, SP 24, IC 47; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	27; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	5	12.6	3.7	50	0.3
8-11.9	13	17.1	9.6	145	0.7
12-15.9	36	25.6	26.7	561	2.8
16-19.9	46	19.8	34.1	866	4.3
20-23.9	49	14.1	36.3	1027	5.2
24-27.9	43	8.7	31.9	965	4.8
28-31.9	38	5.9	28.1	933	4.7
32-35.9	25	3.0	18.5	642	3.2
36-39.9	16	1.5	11.9	489	2.5
40-43.9	11	0.9	8.1	302	1.5
44-47.9	13	0.8	9.6	355	1.8
48-51.9	2	0.1	1.5	59	0.3
52-55.9	2	0.1	1.5	69	0.3
60-63.9	1	0.0	0.7	44	0.2
All	300	110.2	222.2	6509	32.7

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	96	34.7	71.1	19.4	2117	10.6
PP	85	32.9	63.0	18.7	1993	10.0
DF	48	16.5	35.6	19.9	977	4.9
IC	47	15.8	34.8	20.1	918	4.6
SP	24	10.4	17.8	17.7	504	2.5
All	300	110.2	222.2	19.2	6509	32.7

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 111 plots, 300 ft grid (orange: QA flag)

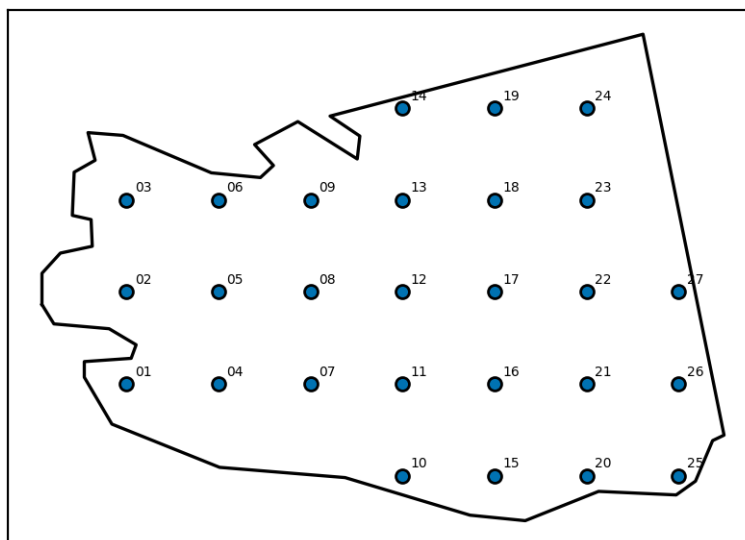


Figure 1. Plot locations with QA status.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
111-01	13	260	26	ok
111-02	11	220	14	ok
111-03	11	220	21	ok
111-04	9	180	21	ok
111-05	14	280	8	ok
111-06	10	200	14	ok
111-07	13	260	8	ok
111-08	9	180	7	ok
111-09	11	220	24	ok
111-10	8	160	29	ok
111-11	13	260	12	ok
111-12	14	280	13	ok
111-13	12	240	24	ok
111-14	10	200	32	ok
111-15	9	180	11	ok
111-16	11	220	17	ok
111-17	13	260	10	ok
111-18	13	260	15	ok
111-19	10	200	25	ok
111-20	9	180	27	ok
111-21	10	200	12	ok
111-22	12	240	17	ok
111-23	12	240	14	ok
111-24	12	240	18	ok
111-25	6	120	30	ok
111-26	11	220	18	ok
111-27	14	280	26	ok